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Case Report

Gradual spontaneous regression of hepatic hemangioma in a woman with surgical menopause: A decade of imaging follow-up [☆]

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ABSTRACT

Hepatic hemangiomas (HHs), the most common benign liver tumors, may grow, remain stable, or rarely regress spontaneously, with regression potentially associated with reduced estrogen levels. We report a 67-year-old woman with a large HH exhibiting gradual regression (~3 mm/year) over a decade of imaging post-surgical menopause in 2010, without hormone replacement therapy. Serial computed tomography (CT) and ultrasound (US) from 2014 to 2025 showed an initial 5.4 cm lesion, hypoechoic on US due to early-stage (F0-F1) metabolic dysfunction-associated steatotic liver disease (MASLD; Fibrosis-4 index <1.3) but with classic CT features (discontinuous peripheral nodular enhancement, centripetal fill-in). By 2025, the lesion regressed in size to ~2 cm, and loss of previously seen regions of enhancement suggesting sclerosing. This case highlights gradual HH regression years after menopause, emphasizes multimodal imaging for MASLD-related US atypia, and supports conservative management for asymptomatic cases.

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Introduction

Hepatic hemangiomas (HHs) are the most common benign liver tumors, with a prevalence of up to 20% in the general population and a female predominance (~4:1) [1,2]. Typically detected incidentally, they were historically considered stable, but recent longitudinal studies have shown that most hepatic hemangiomas tend to enlarge gradually over time, whereas spontaneous regression occurs only in a small minority of cases [3,4]. Some authors define hemangiomas as

giant if ≥ 5 cm, while others define them as giant if ≥ 10 cm [2,5]. While the growth of hepatic hemangiomas is associated with high-estrogen states, their spontaneous regression, potentially linked to decreased estrogen levels, is less studied. [1,6,7,8]. Imaging is critical for diagnosis: typical HHs appear hyperechoic on US in normal livers but may appear hypoechoic in MASLD due to the increased echogenicity of steatotic parenchyma, which can mimic hypoechoic metastases or other focal liver lesions. Therefore, CT or MRI correlation is essential to confirm the classic enhancement pattern of hemangiomas [9,10]. We present a case of a large HH, de-

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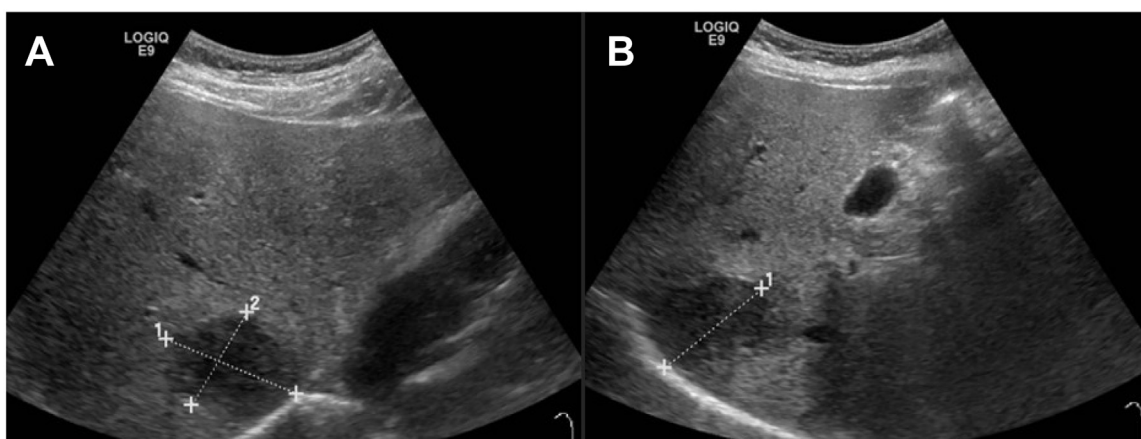


Fig. 1 – Ultrasound (October 2014) of the right hepatic, MASLD background. A, B. B-mode: Hypoechoic lesion (4.9 cm) within diffusely echogenic steatotic liver, atypical for HH due to MASLD.

Table 1 – Timeline of hepatic hemangioma size changes.

Year	US Size (cm)	CT Size (cm)	Notes
2014	4.9	5.4	Initial detection; US hypoechoic due to MASLD; classic CT features (peripheral nodular enhancement, centripetal fill-in); (Fig. 1: US, Fig. 2: CT).
2018	4.6	-	Mild reduction; persistent US hypoechoic.
2019	3.7	-	Stable in size; (Fig. 3).
2020	-	4.5	Near-complete CT fill-in, classic features persisting; (Fig. 4).
2022	3.7	3.8	Involution; reduced CT enhancement; (Fig. 5: CT, Fig. 6: US).
2025	Not visualized	~2	Significant regression; near-undetectable on US, faint residual focus with minimal enhancement on CT; (Fig. 7).

tected four years post-surgical menopause, undergoing significant regression over 11 years of documented imaging suggesting that gradual estrogen decline after oophorectomy and progressive intralesional fibrosis or thrombosis may have contributed to involution [6,11,12].

Case presentation

A 67-year-old woman (age $\approx$ 56 at initial detection) with hypertension, type 2 diabetes, dyslipidemia, and metabolic dysfunction-associated steatotic liver disease (MASLD) [13], who underwent total vaginal hysterectomy with bilateral salpingo-oophorectomy in 2010 (age $\approx$ 52) for uterine prolapse without hormone replacement therapy, had an incidental hepatic lesion detected via abdominal ultrasound (US) in October 2014 during evaluation for mildly elevated bilirubin. Her long-term medications included amlodipine 10 mg once daily, metformin 500 mg twice daily, and atorvastatin 20 mg once daily; she had never received hormone replacement therapy, anti-angiogenic drugs, or other hepatotoxic agents. Her body mass index (BMI) was 24.8 kg/m², classified as overweight by Asian criteria, indicating the potential contribution of adipose-derived estrogen production.

The lesion appeared atypically hypoechoic (4.9 cm maximum diameter) against diffusely hyperechoic steatotic liver parenchyma due to MASLD [9,13] (Fig. 1). In this context, hypoechoic hepatic lesions can mimic metastases, hepatocellular carcinoma (HCC), or focal nodular hyperplasia (FNH); however, stability on follow-up and classic enhancement features on contrast CT favored benign hemangioma.

Contrast-enhanced CT in November 2014 confirmed a giant hepatic hemangioma (4.3 $\times$ 3.5 $\times$ 5.4 cm; maximum diameter 5.4 cm [2,5]) with characteristic features: lobulated hypodense appearance on non-contrast phase, discontinuous peripheral nodular enhancement in the arterial phase, and progressive centripetal fill-in (incomplete in portal venous phase) [10] (Fig. 2). Multiple hepatic and renal cortical cysts were noted.

Serial US and CT from 2014 to 2025 documented gradual shrinkage (Table 1; Figs. 1–7), with the maximum diameter reducing from 5.4 cm (2014 CT) to $\sim$ 2 cm (2025 CT). CT enhancement evolved from robust to minimal (rim-like) by 2025, suggesting possible sclerosing [12]. The lesion remained hypoechoic on US due to persistent MASLD-related steatosis [9,13]. The patient remained asymptomatic, with no complications, and MASLD limited to early-stage steatosis (Fibrosis-4 index $<$ 1.3 [14]). Other findings, such as stable hepatic and renal cysts, were unchanged and incidental.

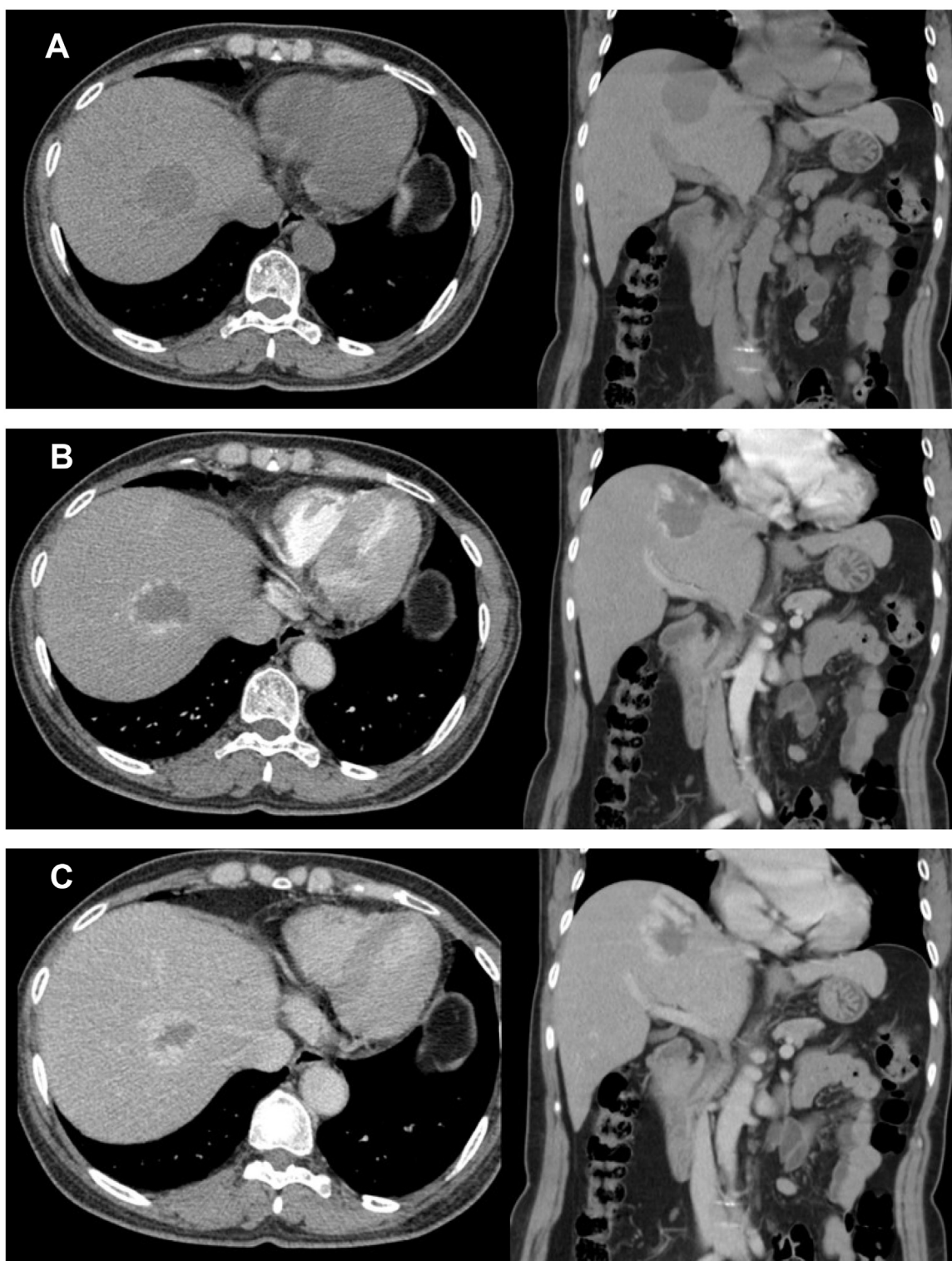


Fig. 2 – Contrast-enhanced CT (November 2014) in the axis/coronal plane. (A). Non-contrast phase: Well-defined hypodense lesion in the right hepatic dome measuring 5.4 cm, with lobulated contour. (B) Arterial phase: Multiple peripheral nodular hyperdense foci compatible with hemangioma enhancement. (C) Portal venous phase: Progressive centripetal fill-in with incomplete fill-in in larger lesions.

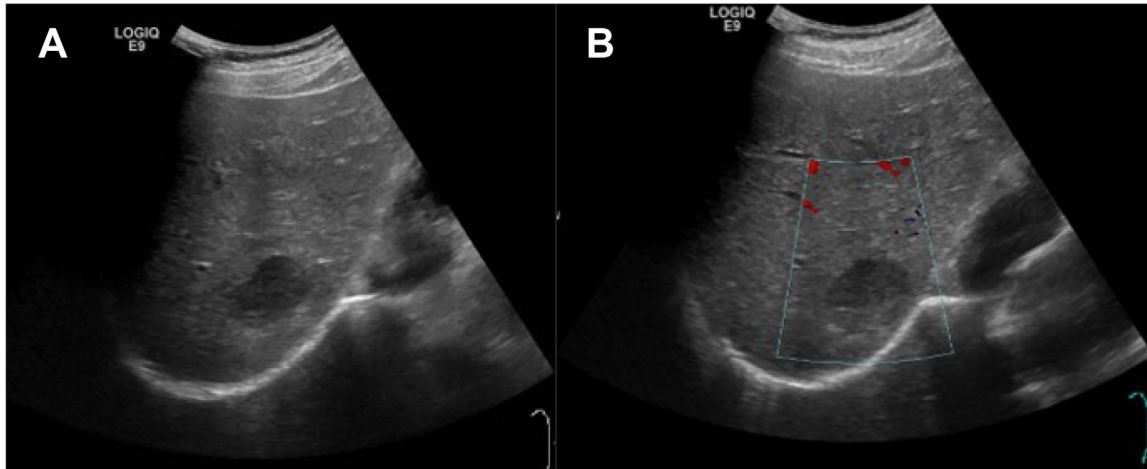


Fig. 3 – Ultrasound (August 2019), right hepatic dome, MASLD background. (A) B-mode: Hypoechoic lesion measuring 3.7 cm within diffusely echogenic steatotic liver; no mass effect on adjacent hepatic vessels. (B) Color Doppler: No internal vascular flow detected.

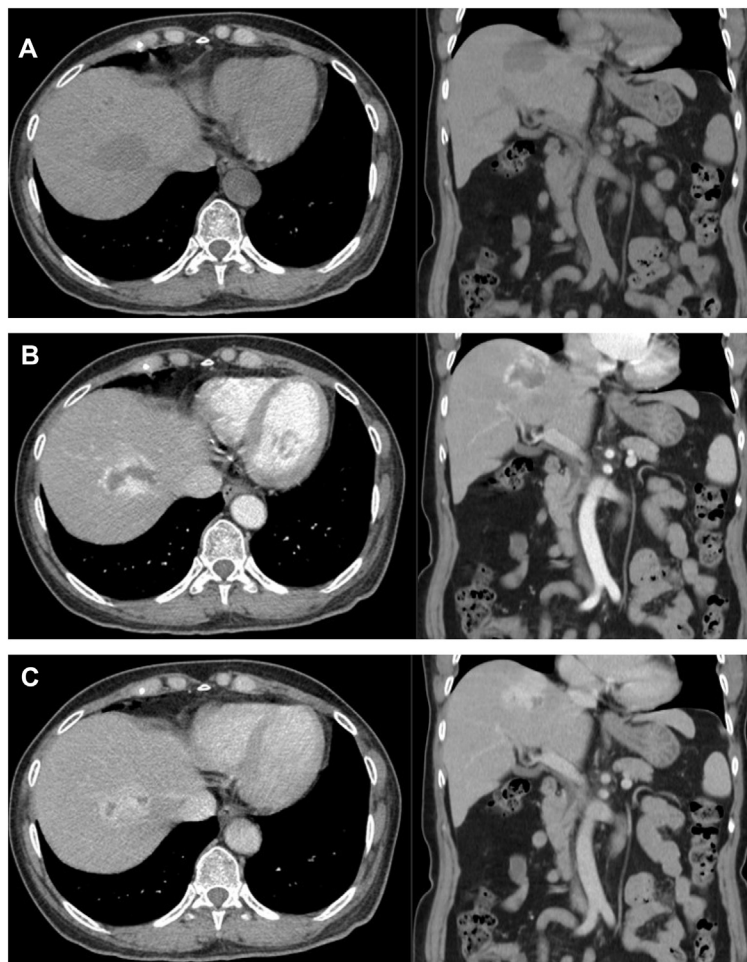


Fig. 4 – Contrast-enhanced CT (November 2020) in axis/coronal plane. (A) Non-contrast phase: Hypodense lesion measuring about 4.5cm. (B) Arterial phase: Peripheral nodular enhancement. (C) Portal venous phase: Near-complete centripetal fill-in, with only a small central hypodense area remaining.

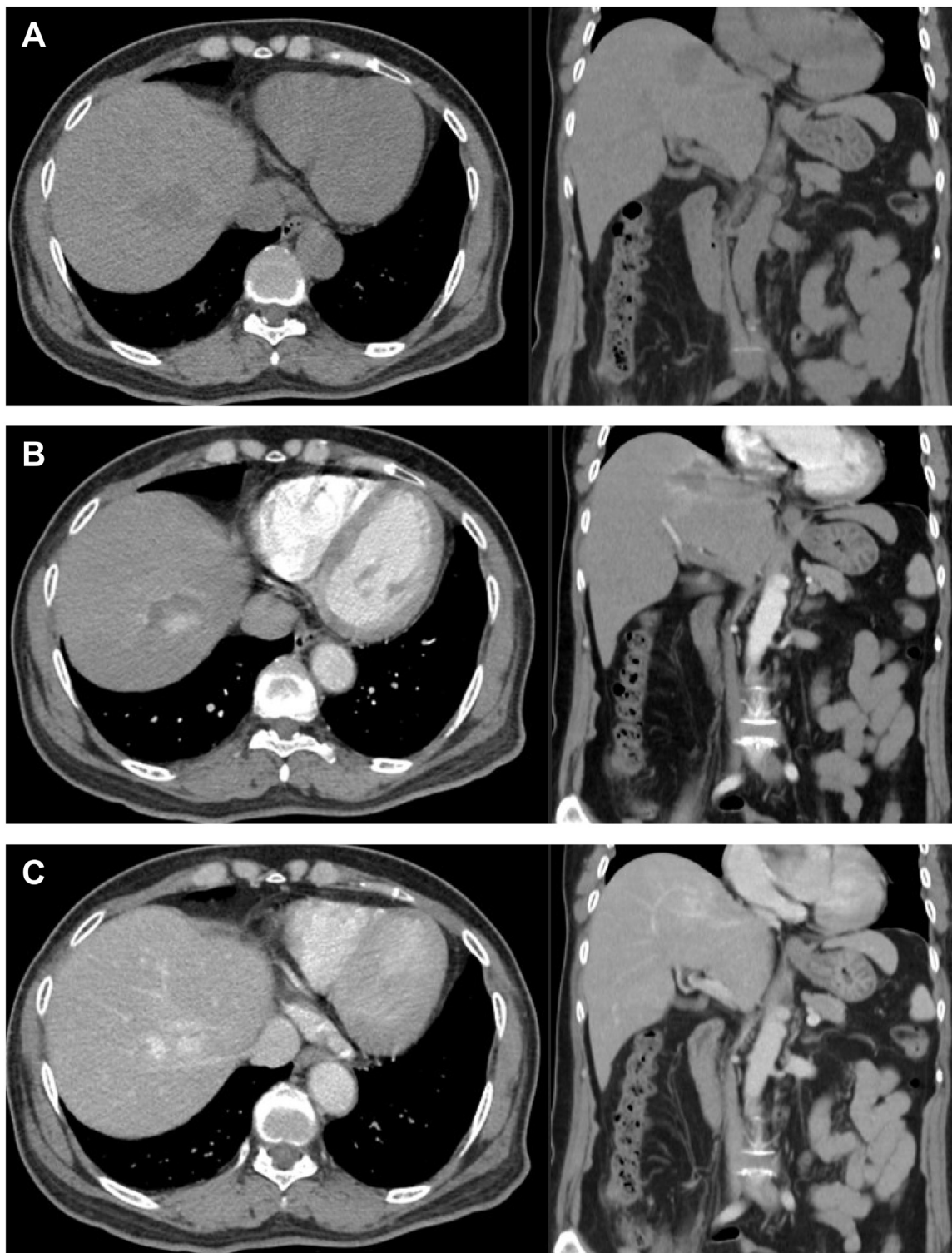


Fig. 5 – Contrast-enhanced CT (May 2022) in axis/coronal plane. (A) Non-contrast phase: Stable hypodense lesion measuring 3.8 cm. (B) Arterial phase: Peripheral nodular enhancement. (C) Portal venous phase: Complete fill-in, with small homogeneous enhancement compared to surrounding liver parenchyma.

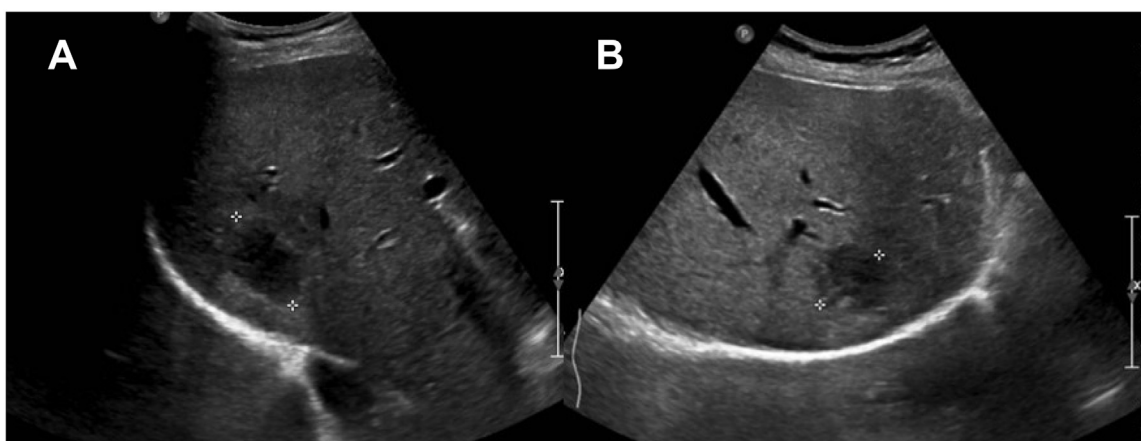


Fig. 6 – Ultrasound (August 2022) of the right hepatic dome in a background of MASLD. (A, B) B-mode: Hypoechoic lesion measuring 3.7 cm within diffusely echogenic steatotic liver; stable appearance with no mass effect on surrounding vasculature.

Maximum diameter reported for consistency. CT used as baseline for reproducibility;

US/CT variations (e.g., 2014: 4.9 cm vs. 5.4 cm) reflect operator differences or MASLD-related hypoechogenicity.

Discussion

Hepatic hemangiomas (HHs) are the most frequent benign hepatic tumors, generally asymptomatic and incidentally discovered. Most HHs remain stable in size, although a minority may enlarge or, less commonly, regress spontaneously [1,3,4,7]. In a prospective series of 236 patients, Li et al. observed that 8.5% of HHs decreased in size during a median 4-year follow-up [3]. Similarly, Aydin et al. documented spontaneous regression in 6.4% of 716 patients [7]. Scarinci et al. also reported a case of giant HH regression in a noncirrhotic adult without intervention, emphasizing the potential for benign natural involution [8]. In our patient, despite limited pre-2014 imaging, the regression rate (~ 3 mm/year) was consistent with the reported trend of size reduction in women over 45 years, particularly among those aged 55–68 (-2.1 ± 1.1 mm/year) [5].

Histologically and radiologically, sclerosed hepatic hemangiomas represent the end stage of cavernous hemangiomas, composed of hyalinized collagenous tissue with scant vascular spaces and abundant elastic fibers [11]. Mori et al. described these lesions as fibrotic, collagen-rich masses with sparse vascular components, which can mimic hepatic malignancies on imaging [11]. Doyle et al. reported that suggestive imaging features of sclerosed hemangioma include a geographic outline, capsular retraction, decrease in size over time, and loss of previously seen regions of enhancement; additional findings can include transient hepatic attenuation difference (THAD), rim enhancement, and nodular regions of intense enhancement similar to typical hemangioma [12]. In our case, serial ultra-

sound (US) demonstrated an atypically hypoechoic lesion in the background of MASLD, contrasting with the classic computed tomography (CT) appearance of discontinuous peripheral nodular enhancement and centripetal fill-in [9,10,13]. The progressive reduction in size and disappearance of previously enhancing regions by 2025 were consistent with sclerosing change typical of hepatic hemangioma. Spontaneous thrombosis and subsequent ischemic fibrosis have also been proposed as mechanisms leading to sclerosis and regression of hemangiomas [11]. These imaging–pathologic findings align with previous reports and highlight the importance of multimodal imaging for accurate characterization, as biopsy was not performed in this case [11,12].

Estrogen promotes endothelial growth and angiogenesis, while its withdrawal reduces these effects and can lead to vascular involution and fibrosis [6,16]. Adipose aromatase is regulated differently by site, sex, and menopause status: in women, cortisol increases aromatase activity in subcutaneous fat, and this effect is stronger in omental fat after menopause [15]. In our overweight patient, residual adipose-derived estrogen may have temporarily supported hemangioma vascularity through pro-angiogenic mechanisms, with later hormonal and metabolic changes favoring involution [15,16]. Ballestri et al. also noted that postmenopausal estrogen deficiency promotes metabolic and vascular remodeling [16]. These hormonal changes may help explain both the hemangioma regression and the presence of MASLD.

Regression can also occur secondary to anti-angiogenic effects. Bevacizumab and sorafenib, both VEGF-pathway inhibitors, have been shown to reduce hemangioma size and alter enhancement patterns [17–19], while azithromycin has been reported to induce regression through possible immunomodulatory or anti-angiogenic mechanisms [20]. Hemodynamic alterations in cirrhosis, leading to reduced hepatic arterial flow, have also been associated with hemangioma regression [7]. However, these mechanisms were unlikely in our patient: her MASLD represented early-stage steatosis (Fibrosis-4 index <1.3), excluding advanced fibrosis or cirrho-

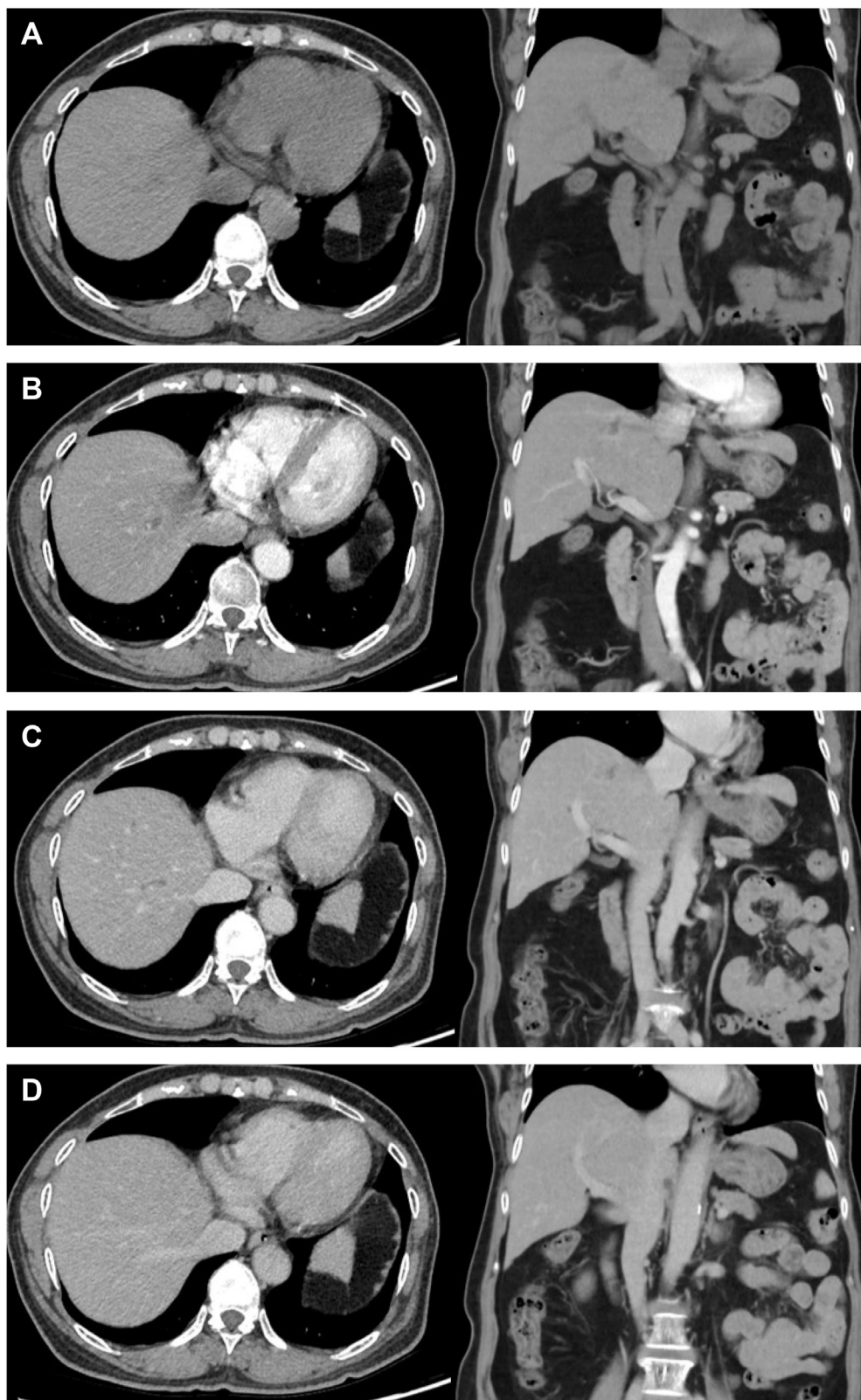


Fig. 7 – Contrast-enhanced CT (July 2025) in axis/coronal plane. (A) Non-contrast phase: Tiny residual hypodense focus in segment VII/VIII (~2 cm). (B) Arterial phase: Small peripheral hyperdense area corresponding to residual vascular component. (C) Portal venous phase: Progressive fill-in, nearly complete. (D) Delayed phase (2 minutes): Complete homogeneous fill-in, attenuation matching adjacent liver parenchyma, confirming the typical hemangioma enhancement pattern.

sis, and she had not received any relevant anti-angiogenic or antibiotic therapies [14,17–20].

Conservative management remains the mainstay for hepatic hemangiomas, as spontaneous regression may occur over time [3,7]. The American College of Gastroenterology recommends observation for asymptomatic lesions, with surgery or intervention reserved for diagnostic uncertainty, symptoms, or complications [2]. Our case shows that even a large hemangioma can regress many years after estrogen withdrawal, highlighting the need to recognize MASLD-related ultrasound changes and confirm stability using classic CT and MRI features [9,10,13].

Conclusion

This case illustrates significant regression of a large hepatic hemangioma over a decade post-estrogen withdrawal. Recognizing atypical US features in MASLD and classic CT and MRI patterns supports noninvasive diagnosis and long-term conservative management. Further studies on long-term imaging in similar cohorts could refine predictors of regression.

Limitation

This single-patient case limits generalizability. Lack of pre-2014 imaging restricts pre- and early post-menopausal insights. US/CT measurement variations may affect accuracy. No MRI was performed, limiting clarity for atypical or sclerosing features. Diagnosis relies on imaging without biopsy.

Patient consent

Written informed consent was obtained for publication of this case report, including the use of clinical data and images.

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